CHAPTER 3

HEAT AND TEMPERATURE



Words like hot and cold are of great importance in our daily life. The temperature of our environment should be such that it allows us to perform our daily work with ease. To keep food fresh the temperature should be low, while to cook it the temperature needs to be high. During hot summer days we need to switch on the fans, while during winters we need blankets. We always like our tea hot, while we enjoy juices as cold. Hence regulating temperature and heat is of much importance for several necessities of our life.

3.1 Hot or cold?

Hot and cold- are these two words sufficient to tell about the temperature of an object? What would you say after touching warm water? It is colder than hot water but warmer than cold water. On keeping hot milk at room temperature you see that it becomes cold after same time. Is it as cold as the milk taken out of a refrigerator?

Just by touching, sensing or seeing we cannot say how much hot or cold an object really is. Our sensitivity towards temperature is not only unreliable but also very limited. Especially for industrial and scientific purpose nobody would like to touch and try to tell the exact temperature of boiling water.

Ordinarily we express temperature comparatively but due to above reasons we had to develop such techniques and instruments by which we could make correct precise and objective measurements for temperature.

3.1.1. Temperature

We know that hot objects can be cooled down while cold objects can be heated too. Can you tell some such examples from your daily life? Discuss amongst yourself.

Analyze the following diagram and state whether hot and cold are two different divisions.

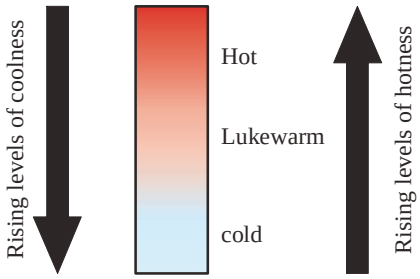


Figure-1: Cold, warm and hot shown on the same scale

As you can see, hot and cold are not two different categories. These two are values on the same measurement scale, one small and another big. We can also say that all objects are hot, some lesser hot while others hotter or it can also be said that all objects are cold, some lesser cold while other colder.

This degree of hotness or coldness is called temperature and its value keeps changing constantly. When we compare the temperature of two objects then the object having higher temperature is called hot and the object with lower temperature is called cold.

3.1.2 How to measure temperature?

To measure the temperature of an object we need following three things.

1. Some property of an object which depends on temperature and which clearly changes with temperature.
2. A material that clearly represents a change in this property and,
3. A universal and commonly used measurement that shows this corresponding change with specific increase or decrease.

If these three conditions are fulfilled then you can make your own thermometer. Thermometers are used in laboratories, hospitals, weather departments etc.

Activity-1: Make your own thermometer

Required material: A small injection bottle, a 10 cm long piece of empty refill, scissors, ink, tap water, candle.

Procedure: Make a small hole in the cap of the bottle using the tip of a scissor. The hole should be just big enough to let the refill pass. Now fill the bottle with tap water up to half and add few drops of ink. Now put the cap on and push the refill inside such that it does not touch the base of bottle. Now close the cap. Completely seal the cap by dropping molten wax from burning candle so that no air can escape from the bottle. Heat your hands by rubbing them together and cover the bottle with your hands (Figure-2). Does the level of water increase in the refill? Why did it happen?

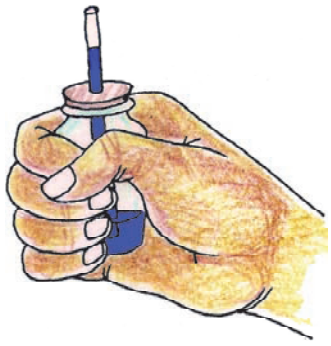
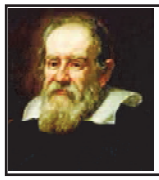


Figure-2: Homemade thermoscope



Galileo is called 'Father of Modern Physics', 'Father of Science' etc. In 1593 he created the first Thermoscope.

This type of temperature measuring device is called as 'Thermoscope' because it does not have a measuring scale as present in thermometer. The change in temperature is shown only through the increase and decrease in the level of liquid filled. It was invented by Galileo. By placing warm hands on the outer surface of a thermoscope, the air inside the bottle gets warmed up and expands. As the bottle is tightly sealed, the warm air exerts pressure on the liquid and forces it to rise up in the straw.

Measuring temperature through volumetric of a liquid expansion is the simplest method for us.

A doctor's thermometer is also based on the same principle. Note down the temperature of hot water, sand, ice etc. using a laboratory thermometer. To understand the structure and method of use of thermometer see instruments used in practical work section given at the end of the book.

3.1.3 Scales of temperature

There are three popular scales for measuring temperature are Fahrenheit,Celsius and Kelvin.

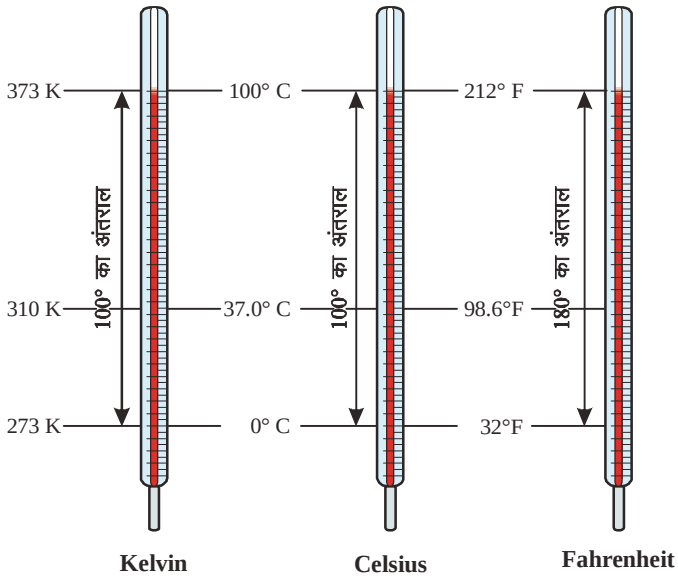
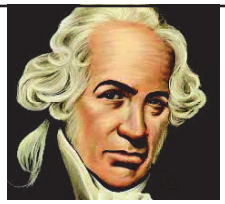


Figure-3: Celsius, Kelvin and Fahrenheit scales



Daniel Gabriel Fahreinheit (1714) is credited with the invention of mercury thermometer.

In Fahrenheit scale– boiling point of water (temperature at which water turns into steam) is 212°F and freezing point of water (temperature at which water turns into ice) is 32°F. This difference in maximum and minimum temperature is divided into 180 similar divisions.

Similarly in Celsius scale boiling point of water is marked at 100°C and freezing point of water is marked at 0°C. Here the difference between the minimum and maximum temperature is divided into 100 similar divisions.

Relation between scales of temperature can be established as following-

$$\left(\frac{\text{Temprature scale - freezing point}}{\text{Boilingpoint - freezingpoint}}\right)\text{of first thermometer} = \left(\frac{\text{Temprature scale - freezing point}}{\text{Boiling point - freezing point}}\right)$$

(eq. 1)

Let us understand this through a solved example-

Example: Our normal body temperature is said to be 98.6° F. What is its value in Celsius?

Solution: $\frac{F-32}{212-32} = \frac{C-0}{100-0}$ = from eq. (1)

$$\Rightarrow \quad \frac{F-32}{180} = \frac{C}{100}$$

$$\Rightarrow \quad \frac{F-32}{9} = \frac{C}{5}$$

$$\Rightarrow \quad \frac{5}{9}\ (F-32) = C$$

Therefore 98.6° F in Celsius is

$$C = \frac{5}{9}\ (98.6–32)°$$

$$C = \frac{5}{9}\ (66.6)°$$

$$C = 37°$$

98.6°F = 37°C is the normal temperature of human body.

In the same way, freezing point of water in Kelvin scale is 273 K at normal pressure and boiling point of water is 373 K. In this scale the minimum and maximum temperature is divided into 100 equal divisions.

Example: If the temperature of a room is 0° K, what is its temperature in Celsius scale.

Solution:

$$\frac{K-273}{373-273} = \frac{C-0}{100-0}$$

$$\frac{K-273}{100} = \frac{C}{100}$$

Therefore K–273 = C ..... or K = C + 273

Hence C = 0 – 273

= –273

0K or –273° C is also called as absolute zero because temperature below this cannot be attained. This is the least temperature that we can possibly achieve.

Think

The temperature of a substance is T. If we divide the substance in 2 equal parts, does the temperature get divided into two? Similarly if the temperature of two equal quantities of substance is T each does the temperature get double when we add both the quantities? Discuss this with your friends.

3.2 Heat

If a small utensil of hot water is kept inside a large utensil of cold water then we see that very soon the hot water gets colder and the cold water gets warmer. After some more time, the temperature of both becomes equal. You must have experienced this in your daily life. To save food from spoiling the utensil is kept inside another utensil containing cold water.

When two objects having two different temperatures are brought in contact then the temperature of each of them increases or decreases until it becomes equal. You must have noticed this. But have you ever thought that why and how this happens? What is the reason behind this increase or decrease in values of temperature?

Earlier it was believed that when two objects at different temperatures are brought together, exchange of an invisible fluid takes place. Quantity of this fluid was assumed to be greater in hotter objects as compared to colder objects. Therefore it was believed that a flow of this fluid takes place from hotter to colder object until both objects have the same amount of this fluid. This fluid was called 'caloric'. People thought that on increasing the quantity of caloric, temperature increases.

How was the theory of 'Caloric' falsified?



Count Rumford

Benjamin Thomson (1753-1814) who was later known as Count Rumford was a renowned scientist and a military advisor.

A special kind of a machine having metal cutters was used to make holes in military cannos. These special cutters were rotated using horse-power. While boring a hole, the machine used to get very hot due to friction. According to the Caloric theory, on boring the caloric inside the cannons would start seeping into the barrels making them hot. But Rumford did not agree with this explanation. He carried out several experiments to test the caloric theory and found that metals could produce heat due to friction without undergoing any change. Production of heat did not affect mass of the metal. He concluded that the movement of horses that provided the power to bore cannons was also the source of heat produced during boring. This means that it was horse power that was converted into heat by friction. In 1798 he proposed that heat could not any material but was a form of energy. Newton had also held a similar view. Joule's experiments in 1840-50 also proved this point. Joule demonstrated an equivalence between mechanical energy and heat energy and you will study more about this in higher classes.

Today we know that heat is not a substance. Rather, it is a form of energy that causes change in temperature. The type of energy which lets us feel the hotness of a object is called heat energy. Heat energy always flows from an object at high temperature to a object at low temperature. On increasing the heat, the temperature of an object increases while on decreasing the heat, its temperature decreases. Hence on adding heat, temperature increases while on removing heat, temperature decreases.

Discuss

Why does an object feel hot or cold upon touch? On touching with your hands what happens to the direction of flow of heat in both the situations?

Units of heat energy

1. In C.G.S. the unit of heat is calorie (Cal). 1 calorie heat is the amount of heat required to raise the temperature of 1 g of water from 14.5°C to 15.5°C. In place of calorie another unit called kilo calorie (kcal) can also be used. 1 kcal = 1000 cal.
2. As heat is a from of energy, therefore the S.I. unit of heat is Joule.

   1 calorie = 4.18 Joule

   1 kilo calorie = 4.18 × 1000 Joule

3.2.1 Transfer of heat

We saw that just as a liquid always flows from higher level to lower level, heat energy also flows from an object at higher temperature to an object at lower temperature. The transfer of heat from one place to another due to difference in temperature is called heat transfer.

In class-9 you have studied about sound and that the transfer of sound always requires a material medium.

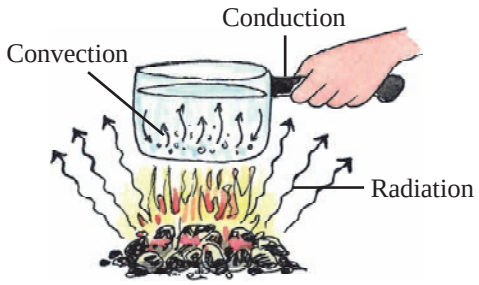


Figure-4: Three types of heat transfer

Does the transfer of heat always require a material medium?

Come, let us know about the three types of heat transfer (figure-4).

1. Conduction

Activity-2

Materials required: Candles, a rod of aluminum, glass & wood rods (each of same length and thickness), awl-pins.

On each rod, stick awl-pins at regular distances using wax work. Tie a piece of cloth at one end of the rod, so that it is easier to hold it on heating. Put the other end on the flame of a candle and heat it. What do you see (figure-5 (a))?

Note down the time at which the awl-pins fall. Do this for every rod. Can you tell from which rod did the awl-pin fall first and why?

Now stick awl-pins on a rod such that the space in middle is empty. Stick the pins at regular gaps on both sides (figure-5 (b)). Now, what happens when you heat the rod from the middle? From which direction do the awl-pins fall first? What can you say about the direction of heat transfer?

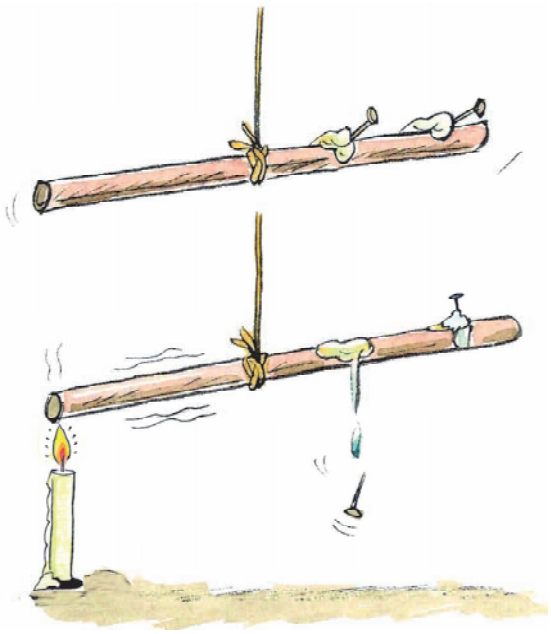


Figure-5(a)

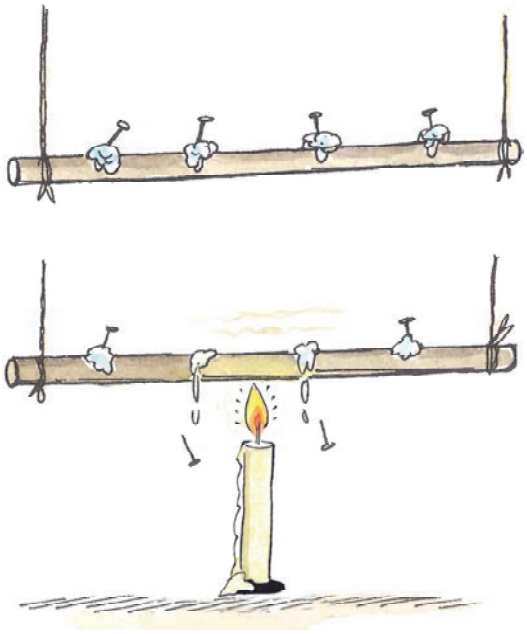


Figure-5(b)

In the process of conduction, heat transfer takes place from one molecule of the substance to another molecule without any actual movement of these molecules. You saw that on heating one end of a metal rod, the other end gets hot very quickly. On heating up, the molecules in that part of the rod start vibrating fast and their heat energy increases. These vibrating molecules pass on the energy to the next molecules. This is the only method by which heat transfer takes place in solids.

Heat transfer through conduction is called heat conductivity. It depends upon the nature of a substance. Based on heat conductivity, substances can be classified into three types-

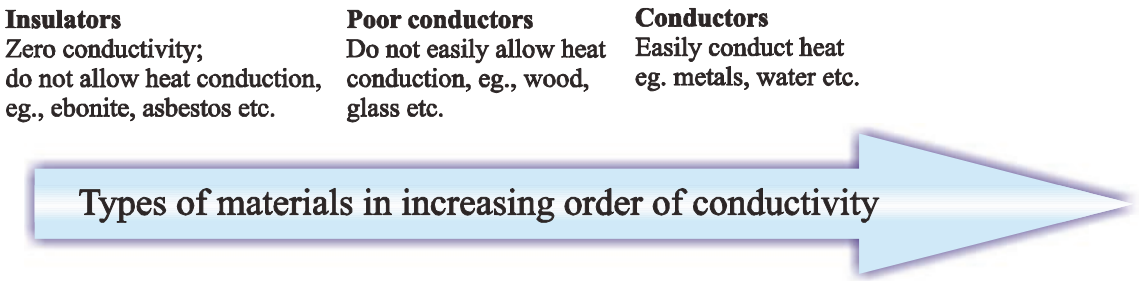


Figure-6: Materials can be insulators, poor conductors or good conductors

2. Convection

Activity-3

Procedure: Fill a flask with water and heat it on a spirit lamp. Add few drops of ink in the flask. Can you see convection currents?

On heating a liquid in a flask, the liquid near the base warms up quickly, becomes lighter and rises up. Its place is taken by the heavier, colder liquid. This process continues till the temperature of the whole liquid is same. This way, convection currents are produced. Heat transfer in gases and liquids take place in the same way.

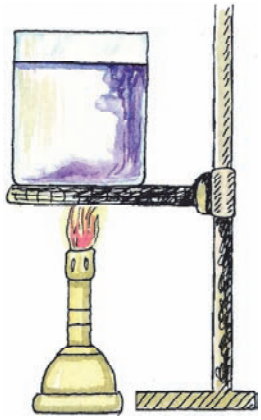


Figure-7: Convection lines will look like this

Question: Can you explain why heat transfer cannot take place by the process of convection in solid substances?

You can observe convection process in several natural phenomena.

The sand on the river coast gets warm during day time.

This results in warming up of the air above the warm sand. This warm air rises up and is replaced by cold breeze coming from the sea. In a similar way, the sand gets colder during night-time while the air above the sea water is warmer. This warm air above sea rises and is replaced by the colder air from the sand. This way, the cycle of land breeze and sea breeze continues.

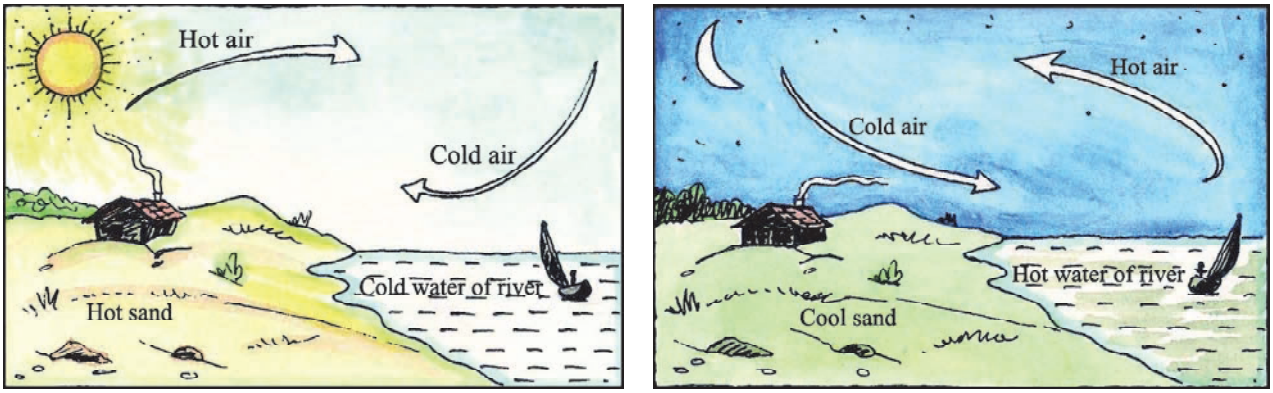


Figure-8: Wind cycle

3. Radiation

Does the transfer of heat energy always require a material medium? Think about the heat obtained through the sun. Air is absent in space, then how do we receive the heat from the sun?

Like light energy, transfer of heat energy can also take place in vacuum. Through radiations, heat travels in a straight line just like light.

Heat is transferred in all directions uniformly. To see this, put your hands around a bulb, lantern or a lamp. You will feel equal warmth in all directions.

When heat energy reaches an object through radiations, then same amount of heat is absorbed by the object and is called as a good absorber. A good absorber of heat is also a good emitter of heat.

Think

Why do we wear light colored cloths during summers and dark colored cloths during winters?

You will find the answer to this question through the following activity.

Activity-4

Materials required: A utensil painted with black color, a similar shiny utensil (of steel) and another earthen utensil.

Procedure: Fill the 3 utensils with water and put them in sunlight. After some time, note down the temperature of water in all three utensils using a thermometer. Which utensil has water with highest temperature and which one has lowest temperature water?

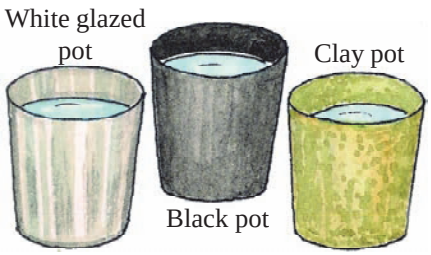


Figure-9: Heat emission and absorption of the object depends on its color and nature.

In the same way put all the three utensils in a cold place. The temperature of which utensil reduces the fastest? You may notice that the absorption and emission of heat depends on two factors (1) on the temperature of object and (2) on the color of the object or its surface (smooth or rough).

3.2.2 Effects of heat

We cannot see the transfer of heat energy is our daily life but we can at least experience its effects. On giving heat, the temperature of an object increases. The increase in temperature depends upon the amount of heat obtained and it leads to expansion of substances. This can also lead to change in the state of substances.

1. Increase in temperature

Have you ever thought that why an iron bench gets hotter than a wooden bench when both are kept under the sun? The sand near a river gets warmer than the river water.

On heating up an object, it absorbs heat and its temperature increases. On cooling down, the object releases heat and its temperature falls. To equally raise the temperature of different substances of same mass, we need to provide different amounts of heat. The quantity of heat (Q) absorbed and emitted by an object depends on the following-

1. On the mass of substance $Q \alpha m$ .....(1)
2. On the change of temperature $Q \alpha \Delta T$ .....(2)
3. On the nature of the material

Hence, $Q \alpha m. \Delta T$

$Q = m.S. \Delta T$ .....(3)

When S is proportionality constant and its value depends upon the nature of the substance. This is known as the specific heat or specific heat capacity of the material.

From equation (3)..... $S = Q/m \Delta T$

S.I. unit of specific heat is Joule/kg/°C.

Therefore, the specific heat capacity of any substance is the amount of heat required to raise the temperature of one unit substance by one degree. The substances with higher specific heat capacity take longer to heat up and to cool down.

Similarly, substances with low specific heat require less time to heat up and to cool down.

Think

What could have happened if all substances had same specific heat capacity? Discuss.

If $m = 1, \Delta T = 1$ Then $S = Q$

Question

- Can you tell why water is used to extinguish a fire? (Note:- Only when the fire is not caused because of electric short circuits).
- If 487.5 J of heat is supplied to raise the temperature of 25 g copper from 25° C to 75° C then what is the specific heat of copper expressed in J/g° C.

Solution: We know that-

$Q = mS.\Delta T$

So, $487.5\ J = (25g.) \times S \times (75°C – 25°C)$

$487.5\ J = 25g \times 50°C \times S$

$$S = \frac{487.5J}{25g \times 50°C} = 0.39\ J/g°C$$

Heat Capacity

The quantity of heat required to raise the temperature of the entire mass of the substance by 1°C is called heat capacity.

If the mass of the substance is 'm' and its specific heat is 'S', then,

Heat capacity = mass × specific heat

= m × S

Unit of heat capacity is J/° C

Equivalent temperature of a mixture

Suppose, you have taken water for bathing in two buckets. One of the buckets contains hot water and the other contains cold water. The temperature of water in one bucket water is 80°C and in the other is 20°C. Now in a third bucket add 1 *l* hot water and 1 *l* cold water. What will be the temperature of this mixture of water?

Here, the specific heat of water in both the cases is same.

Hence $S_1 = S_2 = S$

If the temperature of the mixture is t, then

The heat given out by hot object = $m_1 \times S_1 \times$ fall in temperature in temperature

$= m_1 S_1 (t_1 – t)$

[Mass of hot water is m_1 and its initial temperature is t_1]

Heat absorbed by colder object = $m_2 \times S_2 \times$ rise in temperature = $m_2 S_2 (t – t_2)$

[Mass of cold water is m_2 and its initial temperature is t_2]

According to the law of energy conservation, heat taken by the mixture = heat emitted by the mixture

$m_1 \times S_1 \times (t_1 - t) = m_2 \times S_2 \times (t - t_2)$

$m_1 \times \cancel{S} \times (t_1 - t) = m_2 \times \cancel{S} \times (t - t_2)$ $[S_1 = S_2]$

Hence, equivalent temperature $t = \dfrac{m_1 t_1 + m_2 t_2}{m_1 + m_2}$

$\Rightarrow \dfrac{1\times 80 + 1\times 20}{1+1}$

$\Rightarrow \dfrac{80 + 20}{2}$

$\Rightarrow \dfrac{100}{2}$

$\Rightarrow 50°C$

Therefore, the temperature of this mixture of water will be 50° C.

2. Thermal expansion

You must have noticed that when we pour boiling water in glassware, there is a possibility that the glassware might break. This happens because the internal surface of the glassware expands on receiving heat but outer layer does not.

Discuss

Why are telephone and electrical wires tied loosely? Why is a small gap left between each concrete block when placing them while the making a road? Discuss more such examples.

Generally, on providing heat to a substance, its volume increases and on obtaining heat from a substance, the volume decreases.

Heat expansion in a solid substance

1. Coefficient of linear expansion

The expansion in the unit length of an object, which occurs on raising the temperature by 1°C is called, coefficient of linear expansion. This is denoted by 'α'.

$$\alpha = \frac{\text{Increase in length}}{(\text{initial length} \times \text{increase in temperature})}$$

$$\alpha = \frac{\triangle L}{L \times \triangle T}$$

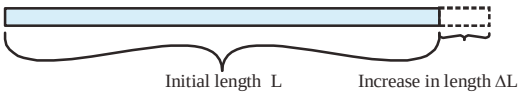


Figure-10(a): Coefficient of linear expansion $\alpha = \dfrac{\triangle L}{L \times \triangle T}$

Hence, $\Delta L = \alpha (L \times \Delta T)$

Increase in length = coefficient of linear expansion × (initial length × increase in temperature)

The unit of thermal expansion is per degree Celcius. Coefficient of linear expansion for iron is 0.00012/° C. Therefore, on raising the temperature by 1°C the length of a 1 m long iron rod increases by 0.00012 m.

2. Coefficient of superficial expansion

The expansion in the area of an object which occurs on raising the temperature of a unit area by 1° C is called coefficient if superficial expansion. This is denoted by 'β'.

$$\beta = \frac{\text{Increase in area}}{\text{initial area} \times \text{increase in temperature}}$$

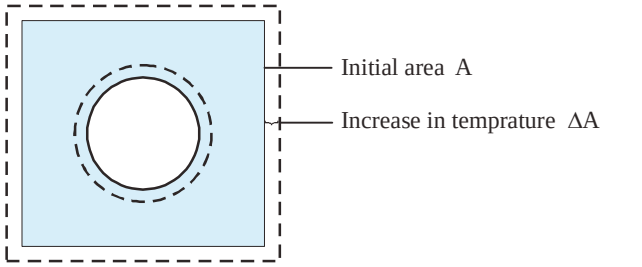


Figure-10(b): coefficient of superficial expansion b = $\frac{\triangle A}{A \times \triangle T}$

$$\beta = \frac{\triangle A}{A \times \triangle T}$$

Hence, $\Delta A = \beta (A \times \Delta T)$

Increase in area = coefficient of superficial expansion × (initial area × increase in temperature)

Unit of β is also /°C. Coefficient of superficial expansion (β) is twice that of the coefficient of linear expansion which means $\beta = 2\alpha$

3. Coefficient of volumetric expansion

The expansion in the volume of a substance, which occurs on increasing the temperature of a unit volume by 1°C, is called coefficient of volumetric expansion. This is denoted by 'γ'.

$$\text{Coefficient of volumetric expansion} = \frac{\text{Increase in volume}}{\text{initial volume} \times \text{increase in temperature}}$$

$$\gamma = \frac{\triangle V}{V \times \triangle T}$$

$$\Delta V = \gamma (V \times \Delta T)$$

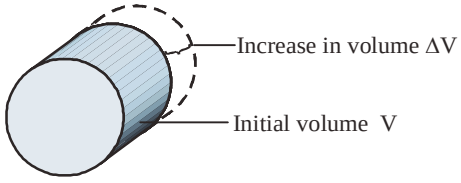


Figure-10(c): coefficient of volumetric expansion $\gamma = \frac{\triangle V}{V \times \triangle T}$

Increase in volume = coefficient of volumetric expansion (initial volume × increase in temperature)

Unit of γ is also /° C. Coefficient of volumetric expansion (γ) is thrice of the coefficient of linear expansion α

Therefore, $\alpha = 2\beta = 3\gamma$

Activity -5: Nail and a loop

Make a loop from a wire such that the head of your nail can easily pass through it. Now heat the head of the nail taken. Can you still easily pass the head through the same loop?

Discuss amongst yourself, why did this happen.

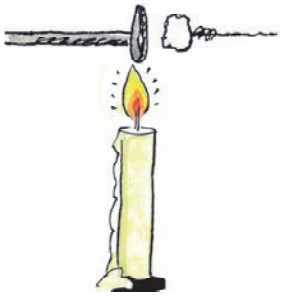


Figure-11: Expansion in volume of solids

Heat Expansion in Liquids

You know that liquids do not have a definite shape. They take the shape of the utensil in which they are kept. When we heat any liquid in a utensil, first expansion of utensil takes place, followed by expansion of the liquid. Hence, volumetric expansion is of two types-

(1) Apparent expansion (2) Real expansion

Come, let us understand the expansion of liquids through the following activity.

Activity-6

Take a glass flask. Put a thin and transparent hollow rod in it. Fill the flask to the brim with water and seal its open mouth with a cork.

Suppose, the level of liquid in the rod is initially at A. On heating the flask, the level of liquid first decreases from A to B and then it increases from B to C, as shown in the figure.

Can you tell why this happens?

On heating, the heat energy first reaches the flask and thus it expands. The level of water decreases. After this, the heat is absorbed by the liquid and it starts to expand. Due to this, the level of water can be seen as rising.

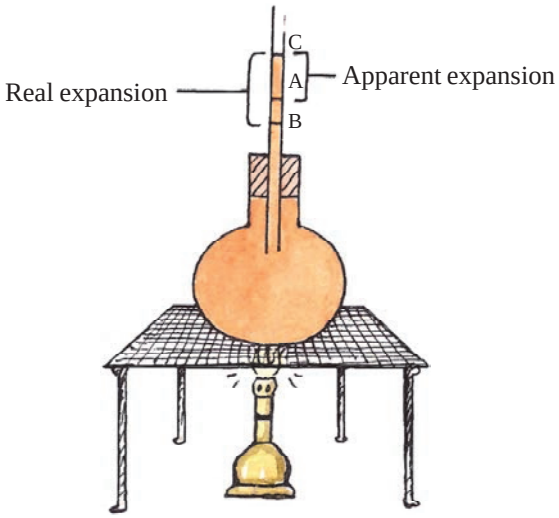


Figure-12: The volume of fluid during thermal expansion

Note down the level at A, B and C. We can see that the apparent expansion of liquid is from A to C, while the real expansion is from B to C. A to B is actually equivalent to the expansion of the flask. For apparent expansion, the expansion of utensil is not taken into consideration. However, in real expansion, the expansion of utensil is also considered.

BC=AC + AB

Therefore, the real expansion of liquid= apparent expansion of liquid + expansion of utensil

Know this: Like all other liquids water also expands on heating. But it is interesting to know that water does not expand between the temperatures of 0°C to 4°C.

This is the reason due to which aquatic life is possible in colder regions. Even if the temperature of uppermost layer of water is less than 0°C, the temperature of lower layers remains 4°C and hence the aquatic life remains safe.

Heat Expansion in gases

Upon heating, the expansion in gases is more than that in liquids and solids. Like liquids, only volumetric expansion is possible in gases. This expansion can be of two types-

1. At constant pressure – If the temperature of a gas is increased keeping pressure constant, then the volume of gas increases. This type of expansion coefficient is called as coefficient of volumetric expansion at constant pressure. On supplying heat to a balloon filled with air, the air inside starts to expand, which can be noticed through the increase in size of the balloon.

Hence, coefficient of volume expansion of a gas $= \dfrac{\text{Increase in volume}}{\text{Initial volume} \times \text{Increase in temprature}}$

This means, $\gamma_p = \dfrac{\triangle \vee}{\vee \triangle T}$

2. At constant volume – if the temperature of gas is increased keeping volume constant, then the pressure of gas increases. This type of expansion coefficient is called as coefficient of volumetric expansion at constant volume. You must have seen a pressure cooker in use. On heating, the whistle of a cooker blows and steam escapes through it. This happens because at constant volume the pressure inside the cooker keeps building up.

Hence, coefficient of pressure expansion of a gas $= \dfrac{\text{Increaseinpressure}}{\text{Initialpressure} \times \text{Increaseintemprature}}$

This means, $\gamma_v = \dfrac{\triangle P}{P \triangle T}$

Activity-7

Take a small empty bottle made of glass and put few drops of water on its mouth. Cover the mouth of this bottle with a coin. Now, rub your hands together and put your palms around the bottle for half a minute. You will notice the coin dancing. Think, why does the coin dance up and down? Which kind of expansion is this, is it an expansion at constant pressure or constant volume?

Question: Why is only volumetric expansion possible in liquids and gases?

3. Change in physical state

On receiving heat, the state of matter may change from one state to another. Study the following figure and understand it using examples.

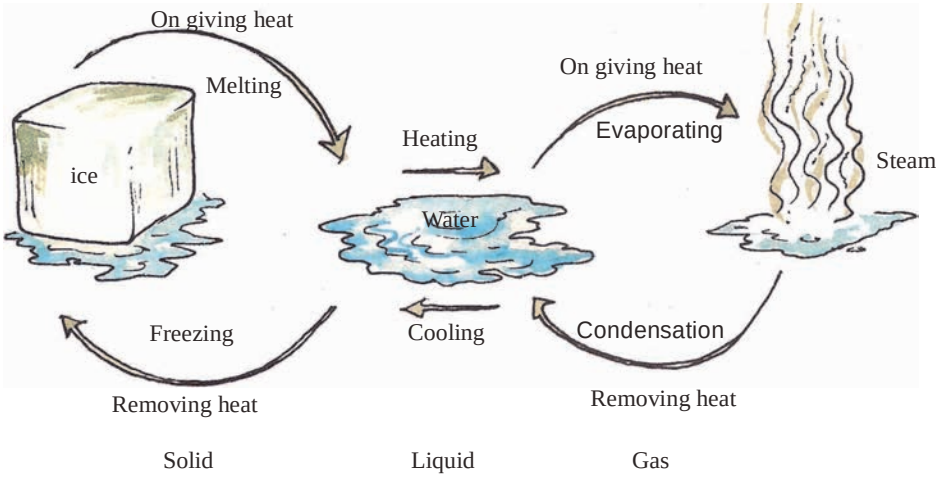


Figure-13: Change in state on providing to or taking heat from a substance.

Activity-8

Take a few ice cubes in a container and note its temperature using a thermometer. As you provide heat to the ice, it starts melting down. Keep noting its temperature until all the ice gets converted into water. Keep heating the water and note its temperature. When the water starts to boil and changes into vapor, note its temperature. Draw a graph of your observatios. This graph would look like the one given in figure-14.

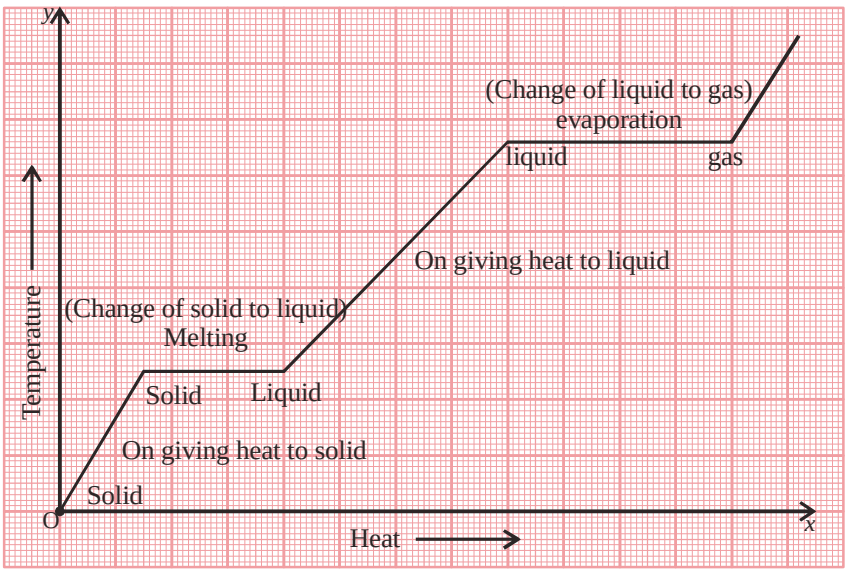


Figure-14: The heat absorbed or given out during state change does not cause increase in temperature.

We saw that on heating, the temperature of a substance keeps increasing. On reaching a certain point in the temperature, the state of the substance changes and temperature becomes constant. This happens because the temperature absorbed during heating process is utilized for state change. As this heat (absorbed or given out) is not evident through rise in temperature, therefore it is called as latent (hidden) heat. Latent heat is of two types.

1. Latent heat of fusion

On providing heat to unit mass of a substance it gets converted from solid state into liquid state. The amount of heat provided during this process is called latent heat of fusion. Latent heat is denoted by 'L'. The unit of latent heat is cal/g or kcal/kg. Latent heat of fusion of ice is 80 kcal/kg. This means that to convert 1 kg ice at 0°C into 1 kg water at the same temperature, requires 80 kcal of heat. If the mass of a substance is 'm' and latent heat is 'L', then the amount of heat given or taken by a substance at a constant temperature is Q=ML. To convert a solid into liquid, same amount of heat is taken away from the substance.

2. Latent heat of vaporization

The amount of heat required to convert a unit mass of water into vapor, at its boiling point, is called latent heat of vaporization. This latent heat is also denoted by 'L'. To change vapor into liquid state, same amount of heat is taken away from the vapor.



What we have learnt

- Hot and cold are measures on the same scale.
- We cannot state the exact temperature of a substance just by touching, seeing or sensing it.
- The degree of hotness or coldness is called temperature.
- It is easy to convert measured temperature from one unit to another.
- Temperature change occurs due to exchange of heat energy.
- S.I. unit of heat is Joule.
- Transfer of heat does not always require a material medium. Through radiation, heat can be transferred even through vacuum.
- In solids, heat transfer can take place only through the method of conduction. Convection can take place only in liquids and gases.
- Heat transfer takes place in each direction uniformly.
- To increase the temperature of equal amount of different substances equally, requires different amounts of heat. This is known as specific heat capacity of a substance.
- In solids, the ratio between linear expansion, superficial expansion and volumetric expansion is 1:2:3.
- In liquids, heat expansion is of two types – apparent expansion and real expansion.
- The heat energy required for change in state is known as latent heat energy.

Keywords

Thermal Energy, Fahrenheit, Kelvin, Calorie, Temperature difference, Heat transfer, conduction, convection, heat capacity, specific heat capacity, coefficient of linear expansion, coefficient of superficial expansion, coefficient of volume expansion, apparent expansion, real expansion, latent heat of fusion and vaporization.

Exercise

1. Choose the correct option-
   (i) Ratio of linear, area and volume expansion in solids is-
   (a) 1:1:1 (b) 1:2:3 (c) 1:2:1 (d) 3:2:1
   (ii) Amongst object A and B, if the specific heat of object A is less than of object B, then-
   (a) Object A will get warm sooner (b) Object B will get warm sooner
   (c) Both will get warm at a same rate (d) None of above
   (iii) Which one amongst the following is best conductor of heat-
   (a) Iron (b) Asbestos (c) Glass (d) Wood
   (iv) In which one of the following heat cannot flow due to convection-
   (a) Tea (b) Water (c) Wind (d) Vacuum
2. Fill in the blanks-
   (i) Heat is not matter instead it is ..... .
   (ii) ..... results in heat transfer.
   (iii) Heat transfer is possible in ..... due to conduction.
   (iv) ..... is not required for heat transfer due to radiation.
   (v) Heat provided to change the physical state of matter results in change in temperature, this process is called ..... .
3. When two objects with different temperatures are put in contact with each other, they both achieve the same temperature after some time. Why is it so?
4. What is latent heat?
5. Balloon filled with gas bursts when we take it close to fire, why?
6. Convert the temperatures given below into given units-
   (i) 14° F into Celsius. (ii) 100° C into Fahrenheit. (iii) 12 K into Celsius.
7. Write three uses of specific heat of water in daily life.
8. Determine the final temperature if we mix one kg of water which is at 40°C with one kg of water which is at 60°C.
9. Mass of an aluminum object is 500 g, how much amount of heat needs to be supplied in order to increase its temperature by 40°C? Specific heat of aluminum is 0.09 J/k°C.
10. Length of an aluminum wire is 100 cm, what will be the increase in its length when its temperature is increased to 50°C from 30°C? Coefficient of linear expansion for aluminum is 26×10-6/°C.
11. Define specific heat capacity.
12. What are the types of heat transfer? Write about them.
13. Give few examples of effects of heat seen in our daily life.